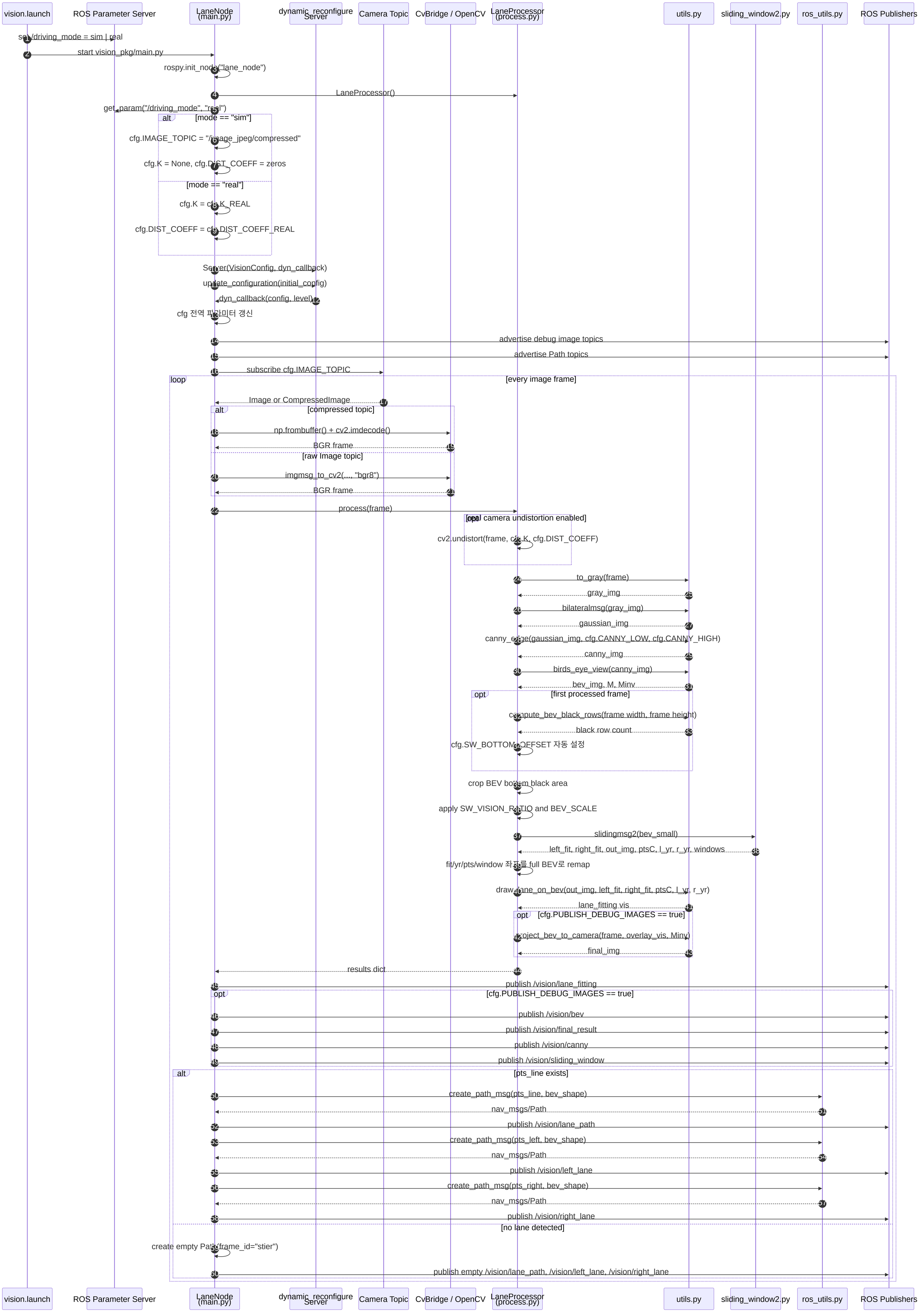
vision.launch
ROS Parameter Server
LaneNode (main.py)
dynamic_reconfigure Server
Camera Topic
CvBridge / OpenCV
LaneProcessor (process.py)
utils.py
sliding_window2.py
ros_utils.py
ROS Publishers
1 set /driving_mode = sim | real
2 start vision_pkg/main.py
3 rospy.init_node("lane_node")
4 LaneProcessor()
5 get_param("/driving_mode", "real")
alt
[mode == "sim"]
6 cfg.IMAGE_TOPIC = "/image_jpeg/compressed"
7 cfg.K = None, cfg.DIST_COEFF = zeros
[mode == "real"]
8 cfg.K = cfg.K_REAL
9 cfg.DIST_COEFF = cfg.DIST_COEFF_REAL
10 Server(VisionConfig, dyn_callback)
11 update_configuration(initial_config)
12 dyn_callback(config, level)
13 cfg 전역 파라미터 갱신
14 advertise debug image topics
15 advertise Path topics
16 subscribe cfg.IMAGE_TOPIC
loop
[every image frame]
17 Image or CompressedImage
alt
[compressed topic]
18 np.frombuffer() + cv2.imdecode()
19 BGR frame
[raw Image topic]
20 imgmsg_to_cv2(..., "bgr8")
21 BGR frame
22 process(frame)
opt
[cal camera undistortion enabled]
23 cv2.undistort(frame, cfg.K, cfg.DIST_COEFF)
24 to_gray(frame)
25 gray_img
26 bilateralmsg(gray_img)
27 gaussian_img
28 canny_edge(gaussian_img, cfg.CANNY_LOW, cfg.CANNY_HIGH)
29 canny_img
30 birds_eye_view(canny_img)
31 bev_img, M, Minv
opt
[first processed frame]
32 compute_bev_black_rows(frame width, frame height)
33 black row count
34 cfg.SW_BOTTOM_OFFSET 자동 설정
35 crop BEV bottom black area
36 apply SW_VISION_RATIO and BEV_SCALE
37 slidingmsg2(bev_small)
38 left_fit, right_fit, out_img, ptsC, l_yr, r_yr, windows
39 fit/yr/pts/window 좌표를 full BEV로 remap
40 draw_lane_on_bev(out_img, left_fit, right_fit, ptsC, l_yr, r_yr)
41 lane_fitting vis
opt
[cfg.PUBLISH_DEBUG_IMAGES == true]
42 project_bev_to_camera(frame, overlay_vis, Minv)
43 final_img
44 results dict
45 publish /vision/lane_fitting
opt
[cfg.PUBLISH_DEBUG_IMAGES == true]
46 publish /vision/bev
47 publish /vision/final_result
48 publish /vision/canny
49 publish /vision/sliding_window
alt
[pts_line exists]
50 create_path_msg(pts_line, bev_shape)
51 nav_msgs/Path
52 publish /vision/lane_path
53 create_path_msg(pts_left, bev_shape)
54 nav_msgs/Path
55 publish /vision/left_lane
56 create_path_msg(pts_right, bev_shape)
57 nav_msgs/Path
58 publish /vision/right_lane
[no lane detected]
59 create empty Path(frame_id="stier")
60 publish empty /vision/lane_path, /vision/left_lane, /vision/right_lane